Model Research Document

Project: AI-Powered Workforce Analytics & Talent Intelligence Dashboard

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Project Statement

This project focuses on developing an AI-Powered Workforce Analytics & Talent Intelligence Dashboard that enables organizations to analyze employee data and make data-driven HR decisions using machine learning, explainable AI, and interactive dashboards.

Dataset Source:

Dataset Name:
IBM HR Analytics Employee Attrition & Performance Dataset

Source:
Kaggle

Models that can be implemented on this dataset

Machine Learning Model	Can be Implemented?	Primary Use Case	Expected Accuracy
Logistic Regression	Yes	Employee Attrition Classification	84–88%
Decision Tree	Yes	Classification	82–87%
Random Forest	Yes	Employee Attrition Prediction	88–93%
Extra Trees Classifier	Yes	Classification	89–94%
Gaussian Naive Bayes	Yes	Classification	82–86%
CatBoost	Yes	HR Analytics Classification	92–97%

Stacking Ensemble	Yes	Ensemble Learning	94–98%
AdaBoost	Yes	Boosting Classification	87–91%
Gradient Boosting	Yes	Classification	89–94%
LightGBM	Yes	Large-scale Classification	91–96%
Support Vector Machine (SVM)	Yes	Binary Classification	86–90%

Models for Regression Tasks

Although the IBM dataset is primarily designed for attrition prediction, some numerical columns can also be treated as regression targets.

Regression Model	Target Variable	Expected Performance
Linear Regression	Monthly Income	Moderate
Random Forest Regressor	Monthly Income	High
XGBoost Regressor	Monthly Income	Very High
LightGBM Regressor	Monthly Income	Very High
CatBoost Regressor	Monthly Income	Excellent
Random Forest Regressor	Years at Company	High
CatBoost Regressor	Salary Hike Prediction	Excellent

Unsupervised Learning Models

Model	Can be Used?	Use Case
K-Means Clustering	Yes	Employee Segmentation
Hierarchical Clustering	Yes	Workforce Grouping
DBSCAN	Yes	Outlier Detection
Gaussian Mixture Model	Yes	Workforce Profiling
Isolation Forest	Yes	Employee Anomaly Detection

Model Evaluation Metrics

The following metrics will be used to compare all Machine Learning models:

- Accuracy
- Precision

- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix
- Cross Validation Score
- SHAP Feature Importance.

Conclusion:

CatBoost is recommended as the primary model, while a Stacking Ensemble provides the highest expected predictive performance. Combined with SHAP and Power BI, the solution delivers accurate, explainable, and scalable workforce analytics.